



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 • STANDARD 6.NOS.6

# Represent Rational Numbers and Their Opposites on the Number Line

Lesson 7-2 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

## My Notes

Level 1 Support



### TODAY'S OBJECTIVES

**Content Objective:** I can place rational numbers, including fractions and decimals, on a number line.

**Language Objective:** I can explain my reasoning using the words rational number, fraction, decimal, and number line.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rational number

Fraction

Decimal

Number line

Equivalent

Integer



Tap any word to see its meaning and a picture.

1

A number you can write as a fraction of two integers, with a bottom number that is not zero, is a .

2

A number that shows part of a whole, like  $\frac{3}{5}$ , is a .

3

A number that uses a point to show parts of a whole is a .

4

A line with numbers in order used to place and compare values is a .

5

Numbers that name the same value, like  $\frac{1}{2}$  and 0.5, are .

6

A whole number or its opposite is an  .

## Write About the Math

**Put the four readings in order from lowest water to highest. How do the decimals let the keeper say something a whole number could not?**

*Pon las cuatro lecturas en orden del agua más baja a la más alta. ¿Cómo le permiten los decimales al encargado decir algo que un número entero no podría?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Rational number**  
*Número racional*

☐ **Fraction**  
*Fracción*

☐ **Decimal**  
*Decimal*

☐ **Number line**  
*Recta numérica*

☐ **Equivalent**  
*Equivalente*

**Model:**  $-1 \frac{1}{2}$  falls between \_\_\_\_ and \_\_\_\_ because \_\_\_\_\_. — Student places  $-1 \frac{1}{2}$  between -1 and -2, halfway, reasoning from the value of the fraction part.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The lowest reading was \_\_\_\_ and the highest was \_\_\_\_.** Tuesday's \_\_\_\_ ft means the water sat  $1\frac{1}{4}$  feet \_\_\_\_ the mark, which is between -1 and \_\_\_\_ on the number line.
  - **My answer is \_\_\_\_ because \_\_\_\_.**  
*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*
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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Rational number
2. **Notes 2:** Fraction
3. **Notes 3:** Decimal
4. **Notes 4:** Number line
5. **Notes 5:** Equivalent
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Rational Numbers on the Number Line is treating the numerator and denominator of a negative fraction as two separate whole numbers instead of one value between two integers — for example, plotting  $-3/4$  by going to  $-3$  and then counting 4 more units to the left, landing (incorrectly) at  $-7$ , instead of recognizing that  $-3/4 = -0.75$ , which sits between 0 and  $-1$ . Before you submit an answer, find the two whole numbers your value falls between, then split that space into equal parts.*
8. **Try It 1:** A.  $-1.5$ ,  $-0.5$ ,  $0.5$  — *The farthest left on the number line comes first:  $-1.5$ , then  $-0.5$ , then  $0.5$ .*
9. **Try It 2:** A.  $-1/4$  —  $-3/4 = -0.75$  and  $-1/4 = -0.25$ . Since  $-0.75$  is farther left (less),  $-3/4$  is to the left of  $-1/4$ .